# CSE 4203 — 4.2 Integer representations + Algorithms

## (*) Addition of Binary numbers

$a = (1110)_2$
$b = (1011)_2$ $\Big\}$ $n = 0, 1, 2, 3$
3 2 1 0

$$a_n + b_n + c_{n-1} = c_n \cdot 2 + s_n$$

$$a_0 + b_0 + 0 = c_0 \cdot 2 + s_0$$
$$0 + 1 + 0 = 0 \cdot 2 + 1$$
$$a_1 + b_1 + c_0 = c_1 \cdot 2 + s_1$$
$$1 + 1 + 0 = 1 \cdot 2 + 0$$
$$a_2 + b_2 + c_1 = c_2 \cdot 2 + s_2$$
$$1 + 0 + 1 = 1 \cdot 2 + 0$$
$$a_3 + b_3 + c_2 = c_3 \cdot 2 + s_3$$
$$1 + 1 + 1 = 1 \cdot 2 + 1$$

R → L

$$a + b = (1001)_2$$

$c_3 \rightarrow s_4 = 1$

$c = 0$
for $i = 0 \rightarrow n-1$
$\quad d = \lfloor (a_i + b_i + c)/2 \rfloor$
$\quad s_i = a_i + b_i + c - 2d$
$\quad c = d$
$s_n = c$

$$d = \lfloor (a_0 + b_0 + c)/2 \rfloor = \lfloor (0+1+0)/2 \rfloor = \lfloor 1/2 \rfloor = 0$$

$$s_0 = 0 + 1 + 0 - 2 \cdot 0 = 1$$

$$c = d$$

## Multiplication Algorithm.

$$ab = a\left(b_0 \cdot 2^0 + b_1 2^1 + b_2 2^2 + \cdots + b_{n-1} 2^{n-1}\right)$$

$$= \underline{a b_0 \cdot 2^0} + a b_1 \cdot 2^1 + a b_2 2^2 + \cdots + a b_{n-1} 2^{n-1}$$

* every time we multiply by 2, shift the bin. expansion 1-digit to the left and add a zero

$a = (110)_2 \qquad b = (101)_2$

$$a b_0 \cdot 2^0 = (110)_2 \cdot 1 \cdot 2^0 = (110)_2$$

$$a b_1 \cdot 2^1 = \underline{(110)_2 \cdot 0} \cdot 2^1 = (0000)_2$$

$$a b_2 \cdot 2^2 = \underline{(110)_2 \cdot 1} \cdot \underline{2^2} = (11000)_2$$

$$\begin{array}{r} 110 \\ 0000 \\ +\,11000 \\ \hline (11110)_2 \end{array}$$

$$\begin{array}{r} 100 \\ \times 120 \\ \hline 000 \\ 200\cancel{\times} \\ 100\times\times \\ \hline 12000 \end{array}$$

# Division algorithm.

$a, d \in \mathbb{Z}$, $d > 0$ then

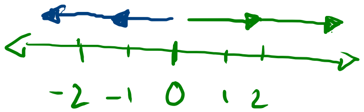
- $q = a \text{ div } d = d | a$
- $r = a \text{ mod } d = a \% d$

$a$ → dividend

$d$ → divisor

↳ $q = 0$

$r = |a|$




while $r \geq d$

$r := r - d$

$q := q + 1$

⇒ if $a < 0$, $r > 0$ then

$r := d - r$

$q := -(q+1)$

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$a = -11 \rightarrow d | a = 3 | -11$ $\quad (-11 - (-12) = 1)$

$d = \underline{3}$

$$\begin{array}{r|r|l} 3 & -11 & -4 \\ & -12 & \\ \hline & 1 & \end{array}$$

↳ $q = 0$

$r = |a| = |-11| = \underline{11}$

$\boxed{\text{while}}$

$r > d$? $11 > 3$ ✓ → $r = 11 - 3 = 8$, $q = 1$

$r > d$? $8 > 3$ ✓ → $r = 8 - 3 = 5$, $q = 2$

$r > d$? $5 > 3$ ✓ → $r = 5 - 3 = 2$, $q = 3$

$r > d$? $2 > 3$ X → exit

$(-11 < 0$ & $2 > 0)$ ✓ →

- $r = d - r = 3 - 2 = 1$
- $q = -(q+1) = -4$